

The `mousik` package*

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Abstract

This is a music-related package focused on writing music scores. It works specially well with small examples, such as fragments in theoretical articles. It is easier than other music writing methods in $\text{\LaTeX}$ because it is based on the `Tikz` package.

Keywords

mousik, music, music theory, music notation, music writing, score, tikz diagram, bar, clef, time signature, note, symbol, rest, dot, chord, accidental, sharp, flat, natural, key

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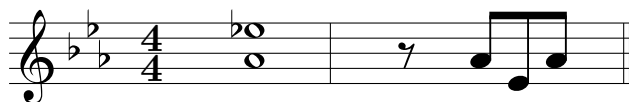
1 Introduction

There are already several ways to set music using Tex. As stated by the ABC package, “*notably MusicTEX and Lilypond*”, as well as ABC per se. Some of them are difficult to use for simple or short fragments; others do not work at all.

The `mousik` package is perfect for short, one-line music fragments. In spite of its compactness, the package is quite complete and versatile, but easy to learn and use.

2 The `mousik` environment

The way to draw a score is by means of the `mousik` environment.



```

\begin{mousik}
\Clef{2}
\Key{-3}
\TimeS{4}{4}

\Note{1}{5}
\Chord
\Acc{-1}{9}
\Note{1}{9}
\Bar

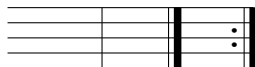
\Rest{8}
\PolyNote{8}{5,2,5}
\FinalBar
\end{mousik}

```

2.1 Ending a line: bars

The only ways to end a line in the score are:

- `\Space`: no bar
- `\Bar`: single bar
- `\FinalBar`: final bar
- `\RepeatBar`: left repeating bar



Any other way to end a line may cause visual errors (see section 4).

2.2 The `\Space` command

The `\Space` command is not only used when no bar is needed. It may also be used to input extra space between symbols. The default space is 2 (which is the default space between symbols in general), but it may be any whole number.

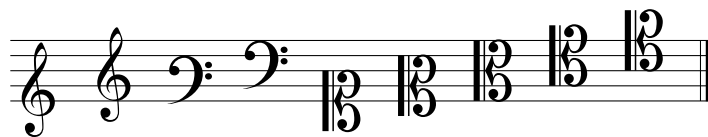


3 Common symbols

3.1 The `\Clef` command

Clefs might be used at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

- `\Clef{1}` and `\Clef{2}` are Treble Clefs in the first and second lines.
- `\Clef{3}` and `\Clef{4}` are Bass Clefs in the third and fourth lines.
- `\Clef{11}`, `\Clef{12}`, `\Clef{13}`, `\Clef{14}` and `\Clef{15}` are Alto Clefs in each of the five lines.



3.2 The `\TimeS` command

Time signatures might be used at any moment on the score. `\TimeS{2}{4}` will draw a 2 by 4 time signature. Any number or symbol can be written inside the brackets.

Also `\TimeS[1]{}{}` will produce a *common time* symbol and `\TimeS[2]{}{}` will produce an *alla breve* symbol.



3.3 Notes and rests

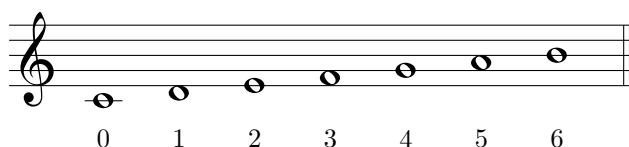
3.3.1 The `\Note` command

The `\Note` command prints a note on the score. It accepts 2 arguments.

The first argument is the duration of the note, from one of these: 1 (whole note / semibreve), 2 (half note / minim), 4 (quarter note / crotchet), 8 (eighth note / quaver), 16 (sixteenth note / semiquaver), 32 (thirty second note / demisemi-quaver).



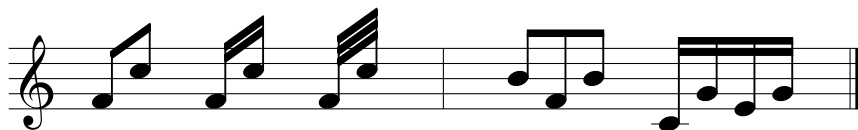
The second argument is the absolute height. These are the notes from 0 to 6:



3.3.2 The `\PolyNote` command

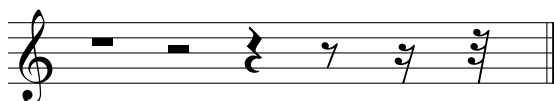
The `\PolyNote` command prints several notes tied together by a beamer. It accepts 2 arguments: the duration of the note and the list of heights separated by commas: `\PolyNote{8}{6,3,6}`.

When there are 2 notes, the beamer is diagonal; when there are more notes, the beamer becomes horizontal.



3.3.3 The `\Rest` command

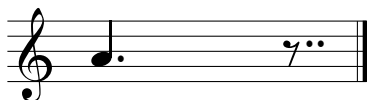
The `\Rest` command prints a rest. It accepts its duration as an argument: 1, 2, 4, 8, 16 or 32.



3.3.4 The `\Dot` command

The `\Dot` command prints any amount of dots in front of a note or rest. It accepts two arguments.

The default number of dots is 1, but it may be any whole number bigger than one. The second argument is compulsory: the height of the dot: `\Dot[2]{7}`.



`\Dot{5}` `\Dot[2]{7}`
`\Note{4}{5}` `\Rest{8}`

3.3.5 The `\Chord` command

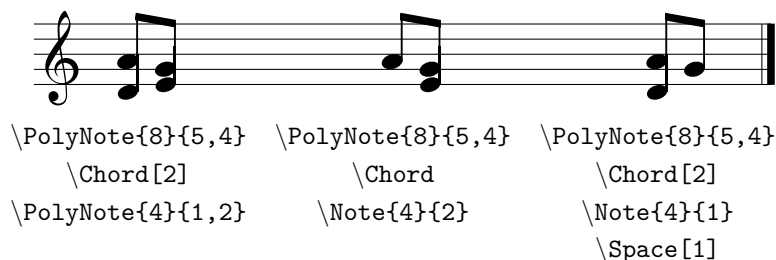
Chords may be obtained by means of the `\Chord` command. It accepts an optional argument: the number of notes to step backwards —mostly for the `\PolyNote` command— with 1 as default.

The best way is to call the `\Note` or `\PolyNote` commands with the highest notes; then call the `\Chord` command; then call the `\Note` or `\PolyNote` commands with the duration 4 of a quarter note.

If the stem needs to be longer, the user may need the `\Enlarge` command, which accepts as arguments the 2 heights of both notes.



For extreme uses of the command, extra `\Space` commands might be needed, as seen here:

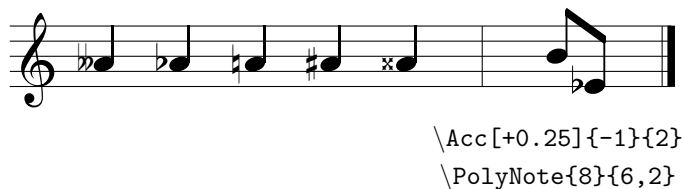


3.4 The `\Acc` command

The `Acc` command prints an accidental before any note. It accepts 2 arguments. The first argument is the type of accidental:

- -2 for the $\flat\flat$ double flat symbol.
- -1 for the $\flat$ flat symbol.
- 0 for the $\natural$ natural symbol.
- 1 for the $\sharp$ sharp symbol.
- 2 for the $\sharp\sharp$ double sharp symbol.

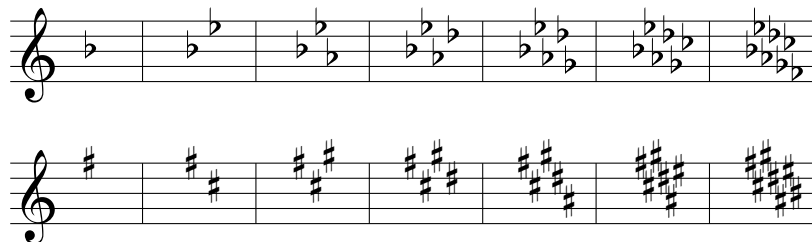
The second argument is the height of the symbol. For usage related to the `\PolyNote` command, an optional argument might be used before the other two, but it is still experimental.



3.5 The `\Key` command

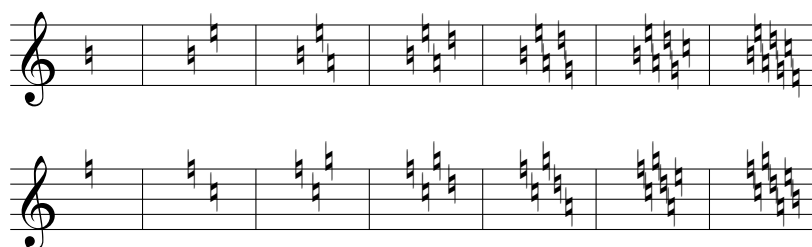
The `\Key` command prints the key at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

It accepts an argument: a positive whole number from 1 to 7 for sharps $\sharp$, and a negative whole number from -1 to -7 for flats $\flat$.



3.6 The `\NoKey` command

The `\NoKey` command is the same as the `\Key` command, but it prints naturals $\natural$ instead of sharps or flats.

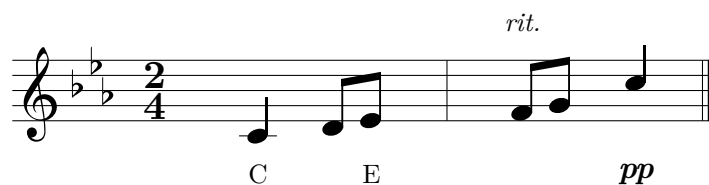


Be careful: the `\Key` and `\NoKey` commands are affected by the current clef! Nonetheless, other commands such as `\Note` or `\Acc` are not affected and need absolute height values.



3.7 The `\Text` command

The `\Text` command allows the printing of any text string, including dynamics and lyrics. It has a compulsory argument, but also an optional one where the user may choose the height of the string: `\Text[2]{\it rit.}`.



4 Common errors

4.1 Overflow of lines



5 The package code

```

1 \RequirePackage[minimal]{leadsheets}
  \useleadsheetslibraries {musicsymbols}
3
4 \RequirePackage{musixtex}
5
6 \RequirePackage{harmony}
7 \RequirePackage{tikz}
8
9 \RequirePackage{xstring}
  \RequirePackage{xparse}
11 \ExplSyntaxOn
    \DeclareExpandableDocumentCommand{\Eval}{m}{\int_eval:n {#1}}
13 \ExplSyntaxOff

```

```

15 \newcounter{modulo}
16 \newcommand{\Modulo}[2]{%
17   \setcounter{modulo}{#1}%
18   \loop%
19     \ifnum\themodulo>\Eval{#2-1}%
20     \setcounter{modulo}{\Eval{\themodulo-#2}}%
21     \repeat%
22     \ifnum\themodulo<0%
23     \setcounter{modulo}{\Eval{\themodulo+#2}}%
24     \repeat%
25   % \themodulo%
26 }
27
28 \newcounter{max}
29 \newcommand{\Max}[2][\themax]{%
30   \setcounter{max}{#1}%
31   \ifnum#2>#1
32     \setcounter{max}{#2}%
33   \fi
34 }
35
36 \newcounter{min}
37 \newcommand{\Min}[2][\themin]{%
38   \setcounter{min}{#1}%
39   \ifnum#2<#1
40     \setcounter{min}{#2}%
41   \fi
42 }
43 \newcounter{first}
44
45 \newcounter{xpos}
46 \newcounter{ypos}
47 \newcounter{clef}
48 \newcounter{key}
49
50 \newcommand{\Staff}{
51   \foreach\i in {0,...,\theypos} {
52     \foreach\j in {0,...,4} {
53       \ifnum\i=\theypos
54         \draw (-0.25,-\i*2+\j*0.2) -- (\thexpos*0.5,-\i*2+\j*0.2);
55       \else
56         \draw (-0.25,-\i*2+\j*0.2) -- (11,-\i*2+\j*0.2);
57       \fi
58     }
59   }
60 }
61
62 \newcommand{\Clef}[1]{
63   \setcounter{clef}{#1}
64   \Adv
65   %
66   \ifnum#1=1
67     \node at (\thexpos*0.5+0.1,-\theypos*2+0.15)
68     {\trebleclef};
69   \fi\ifnum#1=2
70     \node at (\thexpos*0.5+0.1,-\theypos*2+0.35)
71     {\trebleclef};

```

```

73     \fi \ifnum#1=3
        \node at (\thexpos*0.5+0.1,-\theypos*2+0.8)
        {\resizebox{0.7cm}{0.8cm}{\bassclef}};
75     \fi \ifnum#1=4
        \node at (\thexpos*0.5+0.1,-\theypos*2+1)
77     {\resizebox{0.7cm}{0.8cm}{\bassclef}};
        \fi \ifnum#1=11
79     \node at (\thexpos*0.5+0.1,-\theypos*2+0.4)
        {\resizebox{0.65cm}{0.8cm}{\altoclef}};
81     \fi \ifnum#1=12
        \node at (\thexpos*0.5+0.1,-\theypos*2+0.6)
83     {\resizebox{0.65cm}{0.8cm}{\altoclef}};
        \fi \ifnum#1=13
85     \node at (\thexpos*0.5+0.1,-\theypos*2+0.8)
        {\resizebox{0.65cm}{0.8cm}{\altoclef}};
87     \fi \ifnum#1=14
        \node at (\thexpos*0.5+0.1,-\theypos*2+1)
89     {\resizebox{0.65cm}{0.8cm}{\altoclef}};
        \fi \ifnum#1=15
91     \node at (\thexpos*0.5+0.1,-\theypos*2+1.2)
        {\resizebox{0.65cm}{0.8cm}{\altoclef}};
93     \fi
        \addtocounter{xpos}{2}
95 }

97 \input{mousik-key.sty}

99 \newlength{TSLen}
    \newlength{TSHei}
101 \newcommand{\AboveDec}[1]{%
    \settowidth{TSLen}{#1}
103     \settoheight{TSHei}{#1}
    \resizebox{1.5\TSLen}{1.5\TSHei}{#1}
105 }
    \newcommand{\TimeS}[3][0]{
107     \Adv
        \ifnum#1=0
109         \node at (\thexpos*0.5,-\theypos*2+0.6) {\bf \AboveDec{#2}};
        \node at (\thexpos*0.5,-\theypos*2+0.2) {\bf \AboveDec{#3}};
111     \fi \ifnum#1=1
        \node at (\thexpos*0.5,-\theypos*2+0.5) {\meterC};
113     \fi \ifnum#1=2
        \node at (\thexpos*0.5,-\theypos*2+0.575) {\allabreve};
115     \fi
        \addtocounter{xpos}{3}
117 }

119 \newcommand{\Dot}[2][1]{%
    \foreach \d in {1,...,#1}
121     {\node at (\thexpos*0.5+0.1+0.15*\d,-\theypos*2-0.21+0.1*#2)
        {\resizebox{0.1cm}{0.1cm}{\bullet}};
123 }

125 \input{mousik-note.sty}

127 \newcommand{\Text}[2][0]{
    \node at (\thexpos*0.5,-\theypos*2-0.7+#1) {#2};

```

```

129 }

131 \newcommand{\Chord}[1][1]{\addtocounter{xpos}{\Eval{-1-#1}}}
\newcommand{\Space}[1][2]{\Adv\addtocounter{xpos}{#1}}

133 \newcommand{\Bar}{%
135   \ifnum\thexpos=23
     \addtocounter{xpos}{-1}
137   \fi
     \draw (\thexpos*0.5,-\theypos*2+0.8) -- (\thexpos*0.5,-\theypos*2);
139   \addtocounter{xpos}{2}}

141 \newcommand{\FinalBar}{%
     \ifnum\thexpos=23
143   \addtocounter{xpos}{-1}
     \fi
145   \draw[line width=1mm] (\thexpos*0.5,-\theypos*2+0.805) -- (\thexpos*0.5,-\theypos*2-0.005);
     \draw (\thexpos*0.5-0.12,-\theypos*2+0.8) -- (\thexpos*0.5-0.12,-\theypos*2);
147   \addtocounter{xpos}{2}
}

149 \newcommand{\RepeatBar}{%
151   \ifnum\thexpos=23
     \addtocounter{xpos}{-1}
153   \fi
     \node at (\thexpos*0.5-0.25,0.4-\theypos*2) {\resizebox{0.175cm}{0.25cm}{:}};
155   \draw[line width=1mm] (\thexpos*0.5,-\theypos*2+0.805) -- (\thexpos*0.5,-\theypos*2-0.005);
     \draw (\thexpos*0.5-0.12,-\theypos*2+0.8) -- (\thexpos*0.5-0.12,-\theypos*2);
157   \addtocounter{xpos}{2}
}

159 \newcommand{\Adv}{%
161   \ifnum\thexpos>20
     \addtocounter{ypos}{1}
163     \setcounter{xpos}{0}
     \ifnum\theclef=0\else
165     \Clef{\theclef}
     \fi
167     \ifnum\thekey=0\else
     \Key{\thekey}
169     \fi
     \fi
171 }

173 \newenvironment{mousik}
{\noindent\begin{tikzpicture}}
175 {\addtocounter{xpos}{-2}\Staff%
\setcounter{xpos}{0}
\setcounter{ypos}{0}
177 \setcounter{clef}{0}
\setcounter{key}{0}
179 \end{tikzpicture}}

```